Terminal Local Local (2)

```
Windows PowerShell
(C) Корпорация Майкрософт (Microsoft Corporation). Все права защищены.

Попробуйте новую кроссплатформенную оболочку PowerShell (https://aka.ms/pscore6)

(.venv) PS D:\МАГИСТРАТУРА\КУРСЫ\Протоколы_интернета\Киселев Никита\IP_Course_DNS_server> python DNS_Server_prod\test_query_client.py
id 46029
opcode QUERY
rcode NOERROR
;QUESTION
mail.ru. IN A
;ANSWER
mail.ru. 12 IN A 217.69.139.200
mail.ru. 12 IN A 217.69.139.202
mail.ru. 12 IN A 94.100.180.200
mail.ru. 12 IN A 94.100.180.201
;AUTHORITY
;ADDITIONAL
(.venv) PS D:\МАГИСТРАТУРА\КУРСЫ\Протоколы_интернета\Киселев Никита\IP_Course_DNS_server> python DNS_Server_prod\test_query_client.py
opcode QUERY
rcode NOERROR
flags QR RD RA
;QUESTION
e1.ru. IN A
;ANSWER
```

Terminal Local Local (2)

```
e1.ru. 285 IN A 195.19.220.24
;AUTHORITY
;ADDITIONAL
(.venv) PS D:\МАГИСТРАТУРА\КУРСЫ\Протоколы_интернета\Киселев Никита\IP_Course_DNS_server> python DNS_Server_prod\test_query_client.py
id 8347
opcode QUERY
flags QR RD RA
;QUESTION
ya.ru. IN A
;ANSWER
ya.ru. 2 IN A 5.255.255.242
ya.ru. 2 IN A 77.88.55.242
ya.ru. 2 IN A 77.88.44.242
;AUTHORITY
;ADDITIONAL
(.venv) PS D:\МАГИСТРАТУРА\КУРСЫ\Протоколы_интернета\Киселев Никита\IP_Course_DNS_server> python DNS_Server_prod\check_cache.py
JSON представление кэша:
{
  "domain_to_ip": {
    "e1.ru": {
      "195.19.220.24": [
        "2025-06-01T16:22:20.080447",
        "A"
      ]
    },
```

Terminal Local Local (2)


```
    "ya.ru": {
      "77.88.44.242": [
        "2025-06-01T16:17:59.246602",
        "A"
      ],
      "5.255.255.242": [
        "2025-06-01T16:17:59.246602",
        "A"
      ],
      "77.88.55.242": [
        "2025-06-01T16:17:59.246602",
        "A"
      ]
    }
  },
  "ip_to_domain": {
    "195.19.220.24": {
      "e1.ru": [
        "2025-06-01T16:22:20.080447",
        "A"
      ]
    },
    "77.88.44.242": {
      "ya.ru": [
        "2025-06-01T16:17:59.246602",
```

Terminal Local Local (2)

```
        "A"
      ]
    },
    "5.255.255.242": {
      "ya.ru": [
        "2025-06-01T16:17:59.246602",
        "A"
      ]
    },
    "77.88.55.242": {
      "ya.ru": [
        "2025-06-01T16:17:59.246602",
        "A"
      ]
    }
  },
  "domain_to_ns": {},
  "timestamp": "2025-06-01T16:18:25.366999"
}
(.venv) PS D:\МАГИСТРАТУРА\КУРСЫ\Протоколы_интернета\Киселев Никита\IP_Course_DNS_server> python DNS_Server_prod\test_query_client.py
```

IP_Course_DNS_server > M↓ README.md 1:1 LF UTF-8 4 spaces Python 3.12 (pythonProject)

IP_Course-master > M↓ README.md 1:1 LF UTF-8 4 spaces Python 3.12 (pythonProject)